Hridayanshu Raj Acharya

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Profile

AI/ML-focused Computer Engineering student with hands-on experience across the full ML lifecycle and a specialized focus on Generative AI and Agentic AI systems. Proficient in developing RAG pipelines and autonomous agents, alongside data science and model training expertise.

Education

Kathmandu University — Dhulikhel, Kavre
Bachelor of Engineering in Computer Engineering; CGPA: 3.93/4.0 (7 Semesters) — *2022 – 2026*

- **Relevant Courses:** Neural Networks and Deep Learning, Artificial Intelligence, Computer Vision, Statistics and Probability, Calculus and Linear Algebra, Advanced Calculus, Data Structures and Algorithms, Algorithms and Complexity, Software Engineering.

Chelsea World School
High School - NEB, Computer Science, Physics, Chemistry, Maths; GPA: 3.74/4.0 — *2020 - 2022*

Api School
Secondary Education Examination; GPA: 4.0/4.0 — *2017 - 2020*

Experience

Xuno — Kathmandu, Bagmati, Nepal (Hybrid)
Software Developer (AI/ML and Data) - Intern — *Sep 2026 – Present*

Technical Skills

AI/ML: LLMs, RAG, Langchain, LLM Integration, AI Agents, MCP, PyTorch, YOLO, CNN, GNN, Reinforcement Learning, OpenCV, Random Forest
Data Science: NumPy, Pandas, Matplotlib, data cleaning & annotation, feature engineering, model evaluation
Development: React.js, React Native, Node.js, Express.js, Python, C++, Tailwind CSS, JavaScript, HTML/CSS
Databases: MongoDB, MySQL, PostgreSQL, ChromaDB
Tools: Git, GitHub
Languages: Nepali, English, Hindi

AI & ML Projects

Ledgr - Personal Finance & Receipt Intelligence Assistant | *FastAPI, LangChain, React Native*

- Multi-user personal finance SaaS using multimodal LLMs (Gemini Flash) to extract and query financial data from receipts and bank statements.
- Developed a FastAPI-based backend with Pydantic validation and dual-persistence (SQLAlchemy/ChromaDB) for relational and semantic querying.
- Engineered a LangChain agent router to switch between Text-to-SQL for quantitative queries and RAG pipelines for semantic searching.
- Designed as a public multi-user service on Oracle Cloud with secure JWT authentication, a Bring-Your-Own-Key (BYOK) AI pattern, and a cross-platform React Native mobile client.

Multilingual Real-time Hate Speech & Profanity Detection | *XLM-RoBERTa, OpenAI Whisper*

- Real-time English-Nepali hate speech and profanity detection system for virtual meetings and classrooms.
- Built the full ML pipeline (data cleaning, tokenization, multi-class classification) over a 206K+ sentence dataset spanning English, Nepali, and code-switched text across 5 labels.
- Fine-tuned XLM-RoBERTa to 0.777 macro F1 / 96.7% accuracy / 92.7% weighted F1, outperforming DistilBERT and MuRIL baselines.
- Reached per-language macro F1 of 0.78 (English) and 0.74 (Nepali), with class-level F1 up to 0.82 on hate speech.
- Integrated OpenAI Whisper for real-time speech-to-text transcription, enabling live moderation during virtual sessions.

SmartCrimeAI - Multi-Agent Crime Prediction & Patrol Routing | *Random Forest, MARL*

- Interactive multi-agent simulation platform modeling urban crime dynamics to optimize police patrol routing.
- Built tabular (Random Forest) and spatial (Graph Neural Network) crime-prediction models; Random Forest reached 86-88% F1 and 93-94% ROC-AUC across training cycles.
- Designed a multi-agent reinforcement learning (MARL) system for intelligent patrol coordination to cut response times and mitigate crime hotspots.

EyeSpy - Real-Time Exam Cheating Detection | *YOLO, Object Detection*

- AI-powered object detection system to flag cheating behavior in offline exams via live video surveillance.
- Collected and manually annotated 763 HD exam images in YOLO format across cheating / non-cheating classes.
- Trained a YOLO object detection model achieving 0.87 precision / 0.80 recall (non-cheating) and 0.78 F1 (cheating), 0.84 F1 (non-cheating).
- Built the live surveillance frontend to display real-time detection results during exams.

TamangNetra - Trilingual Document Translation | *TMT API, Full-Stack* *Google TMT Hackathon*

- Developed a full-stack document translation platform using the TMT API, implementing semaphore-based concurrency and exponential backoff for rate-limit resiliency.
- Engineered a normalized translation cache and bit-identical reconstruction logic to enable near-instant document generation.
- Ensured high-fidelity trilingual translation (English-Nepali-Tamang) while preserving complex document layouts (PDF, DOCX, CSV, TSV).

Edu Assist - AI-Powered Peer Learning Platform | *RAG, FAISS* *Hackfest 2025*

- Leveraged RAG and FAISS vector search to generate adaptive assessments and personalized learning roadmaps for students.
- Implemented intelligent peer-to-peer tutor matching based on chapter-specific strengths, weaknesses, and meeting preferences.
- Built performance analytics dashboards for tracking subject proficiency and chapter-level weaknesses.
- Managed user authentication and role-based access for students, teachers, and admins.

Certifications

- Google AI Essentials
- Google Data Analysis with Python Specialization